Contact

Yayun Zheng

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Languages

English (Fluent)

Mandarin (Native)

French (Basic)

Research Focus

- Cognitive Psychology
- Human-AI Interaction
- Social Cognition
- Theory of Mind
- NLP & AI Ethics
- Deception Detection

Programming

- R & Python
- NLP (LIWC, VADER)
- SPSS & JASP
- G*Power
- E-Prime
- Qualtrics

Methods Expertise

- SEM
- Mixed-Effects Models
- Network Analysis
- Meta-Analysis
- Sentiment Analysis
- Embedding Clustering

Lab Equipment

EEG (128-channel)

GSR

Geomagic Warp

Eye-tracking

Honors & Awards

Provincial Merit Student (2022-2023)

Provincial Scholarship (2021-2022)

First-Class Scholarships (2020-2022)

Merit Student Award (2021-2022)

Certification

Certified Psychological Counselor, Chinese Academy of Sciences

Yayun Zheng

Cognitive Psychology Researcher | MA Candidate

Research Profile: Cognitive psychology researcher specializing in human-AI interaction, Theory of Mind, and computational approaches to social cognition. Expertise in NLP-based deception detection, cultural bias in large language models, and emotional dynamics modeling. Skilled in experimental design, advanced statistical analysis, and interdisciplinary research bridging psychology, linguistics, and artificial intelligence

Education

Sept 2024 - June 2026, New York University, New York, US
Master of Arts in General Psychology
GPA: 3.94/4.0
Focus: Cognitive Psychology, Human-AI Interaction, Social Cognition
Specializations: NLP applications, AI ethics, computational modeling

Sept 2020 - Jul 2024, Dali University, Yunnan, China
Bachelor of Education in Applied Psychology
GPA: 3.74/4.0
Honors & Awards:
Yunnan Provincial Merit Student (2022-2023)
Yunnan Provincial Scholarship (2021-2022)
School-level First-Class Scholarships (2020-2022)
Merit Student Award (2021-2022)

Research Experience

Fall 2025 (Upcoming), New York University
Head Course Assistant - Introduction to Psychology
Supervisor: Prof. Jay Van Bavel, Department of Psychology

- Appointed as one of two Head Course Assistants, overseeing a large undergraduate course
- Involved in reviewing and coordinating exam materials, including developing protocols to address AI-related academic integrity concerns, organizing in-person exam venues and proposing contingency plans for scheduling conflicts
- Designed and implemented an attendance and participation grading system for recitation sessions
- Assisted in updating the course syllabus and aligning it with key academic deadlines
- Consulted with former Head CAs to incorporate past best practices and manage the team of Recitation Leaders more efficiently

Feb 2025 - Present, New York University
Research Lead - Cultural Bias in ToM Evaluations in LLMs
Department of Psychology

- Authored a first-author research proposal on cultural bias in Theory of Mind (ToM) evaluation in large language models
- Applied psychological theory and cultural frameworks to analyze GPT-4 and DeepSeek performance
- Developed a quantitative coding scheme for benchmark evaluation; drafted manuscript in progress
- Integrated AI ethics and social cognition to explore LLMs' interpretive limitations

Research Experience (Continued)

Apr 2025 - Present, New York University
Researcher Lead - Deception Detection via ToM & Psycholinguistic Features
Department of Psychology

- Designed a two-stage deception framework grounded in Theory of Mind, modeling Type A/B/C reasoning failures with RM and CL cues
- Assembled and annotated a custom multilingual corpus (court transcripts, online reviews, Reddit deception games) for ToM depth and hesitation markers
- Applied LIWC-based features and RoBERTa for deception classification; manuscript in preparation

Mar 2025 - Present, New York University
Researcher - Temporal NLP Modeling of Emotion and Depression Risk on Social Media
Department of Psychology

- Proposed project modeling emotional dynamics (amplitude, inertia, variability) across Reddit, Twitter, and Weibo
- Utilized sentiment tools (LIWC, VADER, GPT-4 API) and linear mixed-effects models
- Planned to build temporal prediction models for depression risk; project at design stage

Oct 2024 - Present, New York University
Research Assistant - Perceptions of AI Creativity and Taste
Department of Psychology, Supervisor: Dr. David Bosch

- Co-designed an IRB-approved study on how users perceive creativity and taste in generative AI art
- Conducted qualitative interviews and mixed-methods analysis on human-AI aesthetic expectations
- Synthesized results into HCI design implications and contributions to social cognition literature

May 2024 - Present, New York University
Lead Researcher - VLTM Activation and Visual Working Memory Capacity

- Proposed and designed a delayed change-detection experiment to examine how activated long-term memory affects visual working memory
- Developed E-Prime protocol; analyzed K-values, reaction time, and accuracy using SPSS and JASP
- Refined theoretical framework under NYU supervision; explored integration with EEG/fMRI methods
- Presented at the 29th Annual NYU MA Psychology Research Conference (2025)

Sept 2023 - May 2024, Dali University, Dali, China
Research Assistant - Meta-Analysis on Pro-Environmental Behaviors
Li Lab

- Conducted an umbrella meta-analysis (n=42 studies) on psychological drivers of environmental behavior (undergraduate thesis)
- Performed systematic literature searches under PRISMA guidelines; extracted and coded effect sizes and moderators
- Applied R packages (metafor, esc) for meta-regression and subgroup analysis
- Visualized findings via forest plots and contributed policy-oriented recommendations

Sept 2021 - Jul 2024, Dali University, Dali, China
Research Assistant - Self-Efficacy & Wellbeing Studies
Li Lab

- Conducted literature reviews on subjective wellbeing, self-regulation, and self-efficacy theories
- Analyzed survey data on Big Five traits and happiness using SPSS, JASP, and psychological network analysis
- Built and interpreted structural equation models; regularly presented progress to lab advisors for feedback and iteration

August 2025, New York University
Foundations of Research Computing (FORC) Camp - Python Track

- Selected for NYU's intensive 3-day training on Python-based data workflows for humanities and social science research

- Trained for word embedding extraction, semantic motif analysis, and Flask-based automation using text corpora
- Applied acquired tools to projects on stereotype learning and emotional modeling in social media text

Publications & Conference Presentations

Conference Acceptance (Forthcoming):

- Zheng, Y. (2026). A theory-of-mind driven error taxonomy for deception detection in courtroom testimonies. Paper accepted at the 2026 Applied Linguistics Conference.

Journal Publications:

- Zheng, Y. (2022). A theoretical analysis of Satir iceberg model for two-child families. *International Journal of Education and Humanities*. https://doi.org/10.54097/ijeh.v4i2.1613
  - Collected literature about family education and Satir's iceberg model and led the project team in desk research
  - Developed visual model to conceptualize and analyzed the two-sibling situation with Satir's iceberg theory

Conference Proceedings:

- Zheng, Y. (2023). The unusual yellow hue and 65% saturation: The interaction of color hue and saturation on time perception. In *Proceedings of the 2023 International Conference on Applied Psychology and Education*. https://doi.org/10.2991/978-2-38476-158-6_35
  - Designed the stimulus-keystroke task with E-Prime psycho software to test subjects' response speed to color stimulus
  - Performed data filtering analysis including elimination of outliers, data merging and recording
  - Presented the study at the 2023 International Conference on Applied Psychology and Education
- Zheng, Y. (2023). How culture affects mental representations of diagonal time lines: Effect of the domain lateral and vertical MTL. In *Proceedings of the 2nd International Conference on Public Culture and Social Services*. https://doi.org/10.2991/978-2-38476-130-2_17
  - Led the literature review and supported team in building theoretical framework
  - Assisted group members with experimental flow charts and other diagrams

Practical Experience

Mar 2023 - Mar 2024, Dali University Mental Health Center, Dali, China
Psychological Counselor

- Provided supervised counseling to students; attended regular case supervision
- Applied evidence-based therapeutic techniques to support student mental health

Oct 2023 - Dec 2023, China National Institute of Standardization (CNIS)
Lab Assistant - Human Factors Research

- Operated EEG and physiological sensors in human-product interaction studies
- Processed and analyzed 3D hand modeling data for national "digital human" project

Jul 2022 - Aug 2022, Haier HR Group
HR Business Partner Intern

- Supported onboarding and psychological training; analyzed hiring metrics and performance data
- Applied psychological principles to organizational behavior and talent management

Sept 2021 - Oct 2023, Dali University Psychology Lab, Dali, China
Lab Assistant - Experimental Psychology

- Assisted in experiment design and data collection using E-Prime and physiological tools
- Maintained lab equipment and managed participant recruitment

Research Tools & Technical Skills

Programming & Statistical Analysis:
R (metafor, esc, ggplot2) ○ Python (pandas, numpy, scikit-learn) ○ SPSS ○ JASP ○ G*Power

Natural Language Processing:
LIWC (Linguistic Inquiry and Word Count) ○ VADER Sentiment Analysis ○ GPT-4 API Integration ○ RoBERTa ○ Word Embedding Extraction ○ Semantic Motif Analysis ○ Collocation Analysis

Experimental Tools & Equipment:
E-Prime (experimental design and data collection) ○ EEG (128-channel) ○ Qualtrics ○ GSR (Galvanic Skin Response) ○ Geomagic Warp ○ Eye-tracking systems

Advanced Modeling & Data Science Methods:
Structural Equation Modeling (SEM) ○ Linear Mixed-Effects Modeling ○ Psychological Network Analysis ○ Meta-Analysis & Meta-Regression ○ Sentiment Analysis ○ Embedding Clustering ○ Text Mining

Research Design & Methodology:
Experimental Design ○ IRB Protocol Development ○ PRISMA Guidelines ○ Qualitative Interviews ○ Mixed-Methods Analysis ○ Corpus Annotation ○ Effect Size Calculation

Specialized Software:
Flask-based Automation ○ Canvas Learning Platform ○ Data Visualization (forest plots, network graphs)

Research Interests

Cognitive Psychology & Human-AI Interaction:
Theory of Mind in humans and AI systems ○ Cultural bias in AI evaluation ○ Human perception of AI creativity and aesthetics ○ Visual working memory and long-term memory interaction ○ Cognitive processes in deception detection

Computational Social Cognition:
NLP applications in psychological research ○ Deception detection using psycholinguistic features ○ Emotional dynamics modeling on social media ○ Temporal prediction of mental health risk ○ Cross-cultural cognition and language

Applied Psychology:
Environmental psychology and pro-environmental behaviors ○ Subjective wellbeing and self-efficacy ○ Psychological counseling and mental health ○ Human factors and ergonomics ○ Time perception and color psychology

AI Ethics & Society:
Algorithmic bias in language models ○ AI academic integrity concerns ○ Ethical implications of AI in human judgment ○ Cross-modal content analysis

Key Competencies

Research Leadership: Lead researcher on multiple independent projects ○ First-author manuscript development ○ IRB protocol design ○ Team coordination and mentorship

Interdisciplinary Integration: Bridging psychology, linguistics, and computer science ○ Applying AI tools to psychological questions ○ Translating theoretical frameworks into computational models

Communication & Teaching: Course assistance and curriculum development ○ Academic integrity policy development ○ Student mentorship ○ Conference presentation experience

Language Proficiency: Native Mandarin speaker ○ Fluent English (academic and professional) ○ Basic French ○ Cross-cultural research capability

Professional Development & Training

- **Foundations of Research Computing (FORC) Camp** - NYU intensive Python training for humanities and social science research (August 2025)
- **Certified Psychological Counselor** - Institute of Psychology, Chinese Academy of Sciences (China)
- **Conference Presenter** - Multiple international conferences in applied psychology and cognitive science
- **Graduate Teaching Assistant Training** - NYU Department of Psychology